Solution: Problem Challenge 2: Rearrange a LinkedList

Problem Statement

Given the head of a Singly LinkedList, write a method to modify the LinkedList such that the **nodes from the second half of the LinkedList are inserted alternately to the nodes from the first half in reverse order**. So if the LinkedList has nodes `1 -> 2 -> 3 -> 4 -> 5 -> 6 -> null`, your method should return `1 -> 6 -> 2 -> 5 -> 3 -> 4 -> null`.

Your algorithm should use only constant space the input LinkedList should be modified in-place.

Example 1:

```
Input: 2 -> 4 -> 6 -> 8 -> 10 -> 12 -> null
Output: 2 -> 12 -> 4 -> 10 -> 6 -> 8 -> null
```

Example 2:

```
Input: 2 -> 4 -> 6 -> 8 -> 10 -> null
Output: 2 -> 10 -> 4 -> 8 -> 6 -> null
```

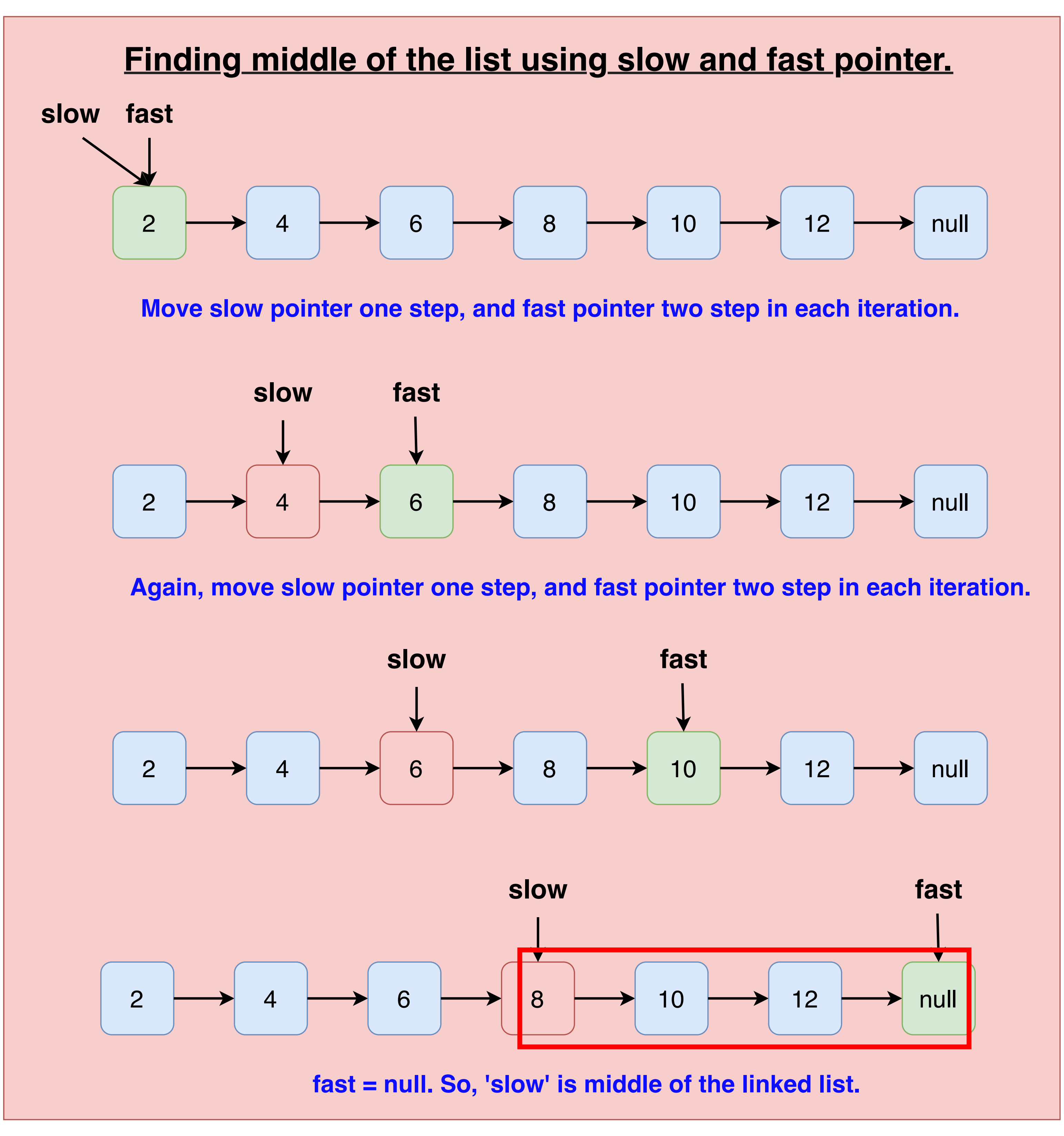
Solution

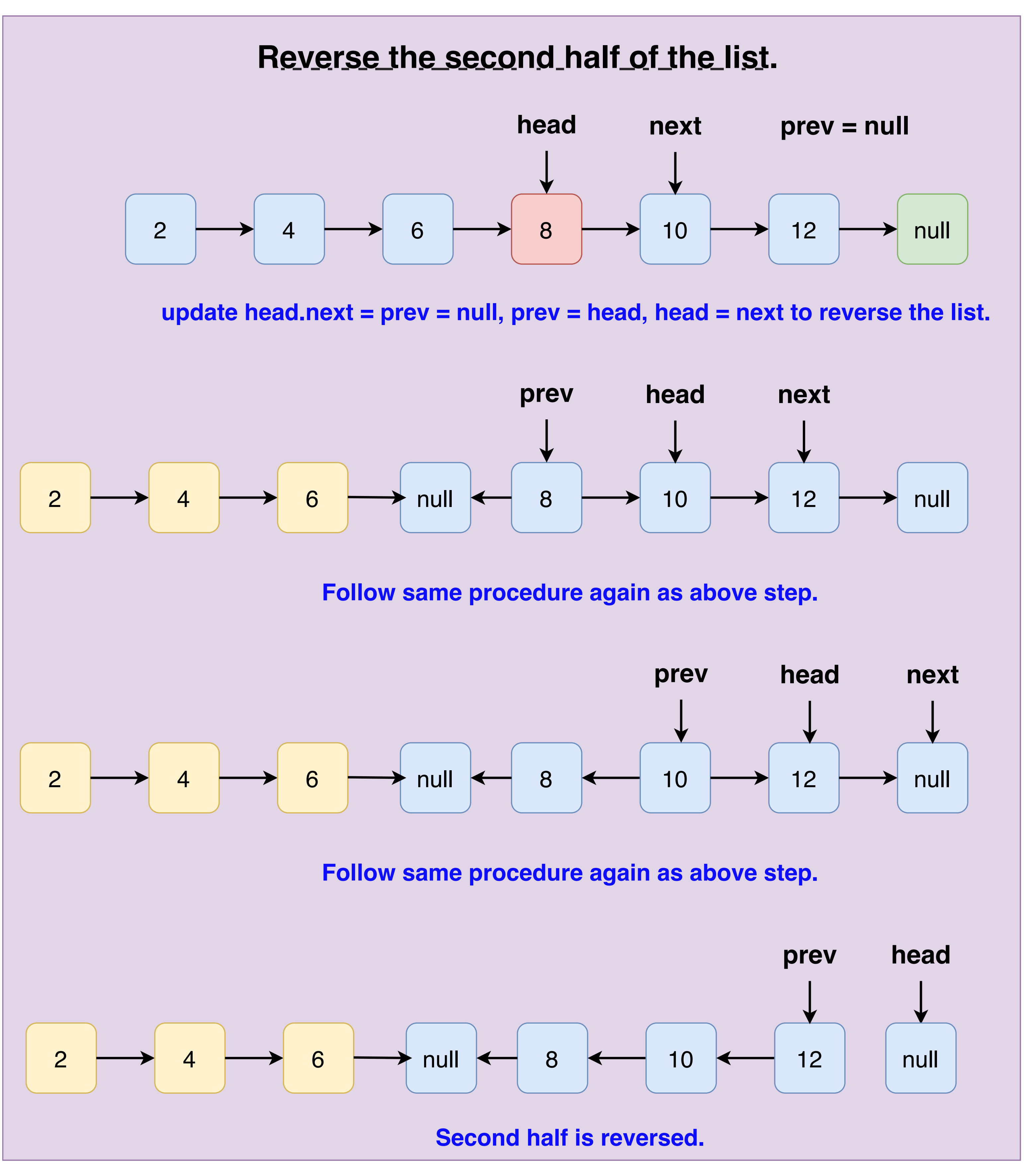
This problem shares similarities with `Palindrome LinkedList`. To rearrange the given LinkedList we will follow the following steps:

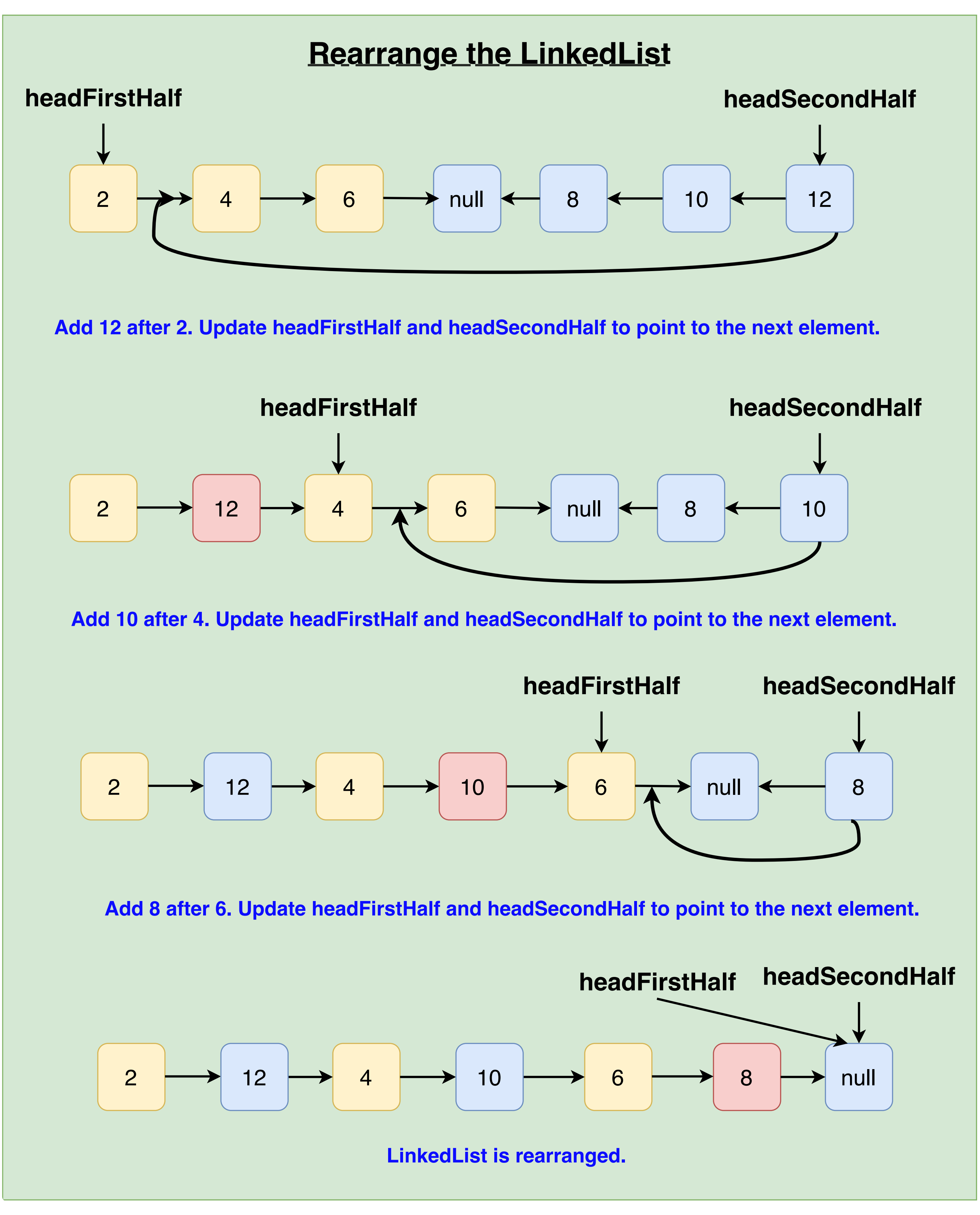
1. We can use the **Fast & Slow pointers** method similar to `Middle of the LinkedList` to find the middle node of the LinkedList.
2. Once we have the middle of the LinkedList, we will reverse the second half of the LinkedList.
3. Finally, we'll iterate through the first half and the reversed second half to produce a LinkedList in the required order.

Here is the visual representation of this algorithm for Example-1:

Three steps to rearranging the LinkedList:
1. Find the middle of the list
2. Reverse the second half
3. Rearrange the list.

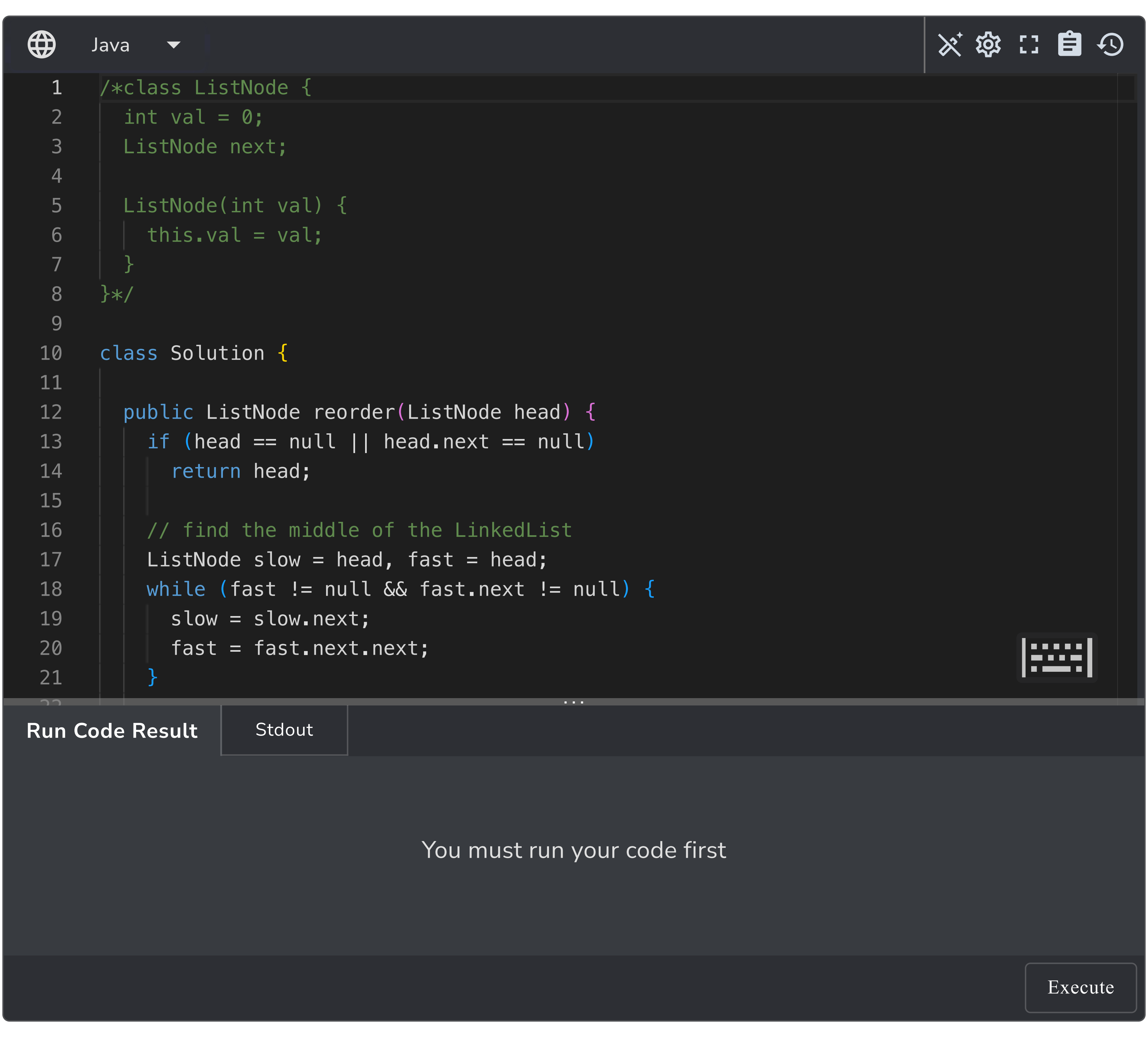

Finding middle of the list using slow and fast pointer.
slow
fast
2
4
6
8
10
12
null
Move slow pointer one step, and fast pointer two step in each iteration.
Again, move slow pointer one step, and fast pointer two step in each iteration.
fast = null. So, 'slow' is middle of the linked list.


Reverse the second half of the list.
head
next
prev = null
prev
2
4
6
8
10
12
null
update head.next = prev = null, prev = head, head = next to reverse the list.
Follow same procedure again as above step.
Follow same procedure again as above step.
Second half is reversed.


Rearrange the LinkedList
headFirstHalf
headSecondHalf
2
4
6
null
8
10
12
Add 12 after 2. Update headFirstHalf and headSecondHalf to point to the next element.
Add 10 after 4. Update headFirstHalf and headSecondHalf to point to the next element.
Add 8 after 6. Update headFirstHalf and headSecondHalf to point to the next element.
LinkedList is rearranged.

Code

Here is what our algorithm will look like:



```java
/*class ListNode {
  int val = 0;
  ListNode next;

  ListNode(int val) {
    this.val = val;
  }
}*/

class Solution {

  public ListNode reorder(ListNode head) {
    if (head == null || head.next == null)
      return head;

    // find the middle of the LinkedList
    ListNode slow = head, fast = head;
    while (fast != null && fast.next != null) {
      slow = slow.next;
      fast = fast.next.next;
    }
```

You must run your code first

Time Complexity

The above algorithm will have a time complexity of $O(N)$ where N is the number of nodes in the LinkedList.

Space Complexity

The algorithm runs in constant space $O(1)$.

← Previous
Problem Challenge 2: Rearrange A LinkedList (Medium)

Next →
Problem Challenge 3: Cycle In A Circular Array (Hard)

☐ Mark as Completed